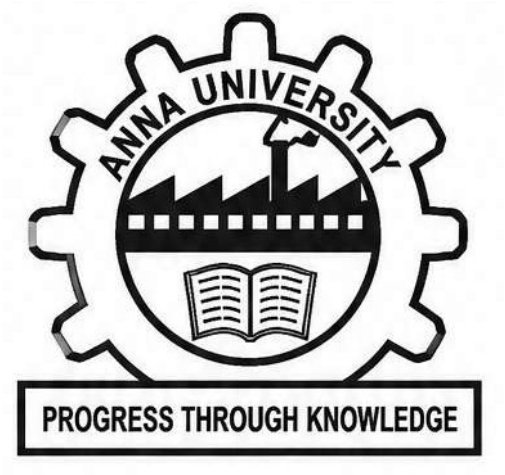
SALES DATA ANALYSIS

A SUMMER INTERNSHIP REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that Summer Internship project report "**Sales Data Analysis**" is the Bonafide work of **HARIKRISHNAN T (810023205039)** has satisfactory completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year **2025-2026**.

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ABSTRACT

The **Sales Data Analysis** project focuses on analyzing sales data to understand business performance and support effective decision-making. The project provides a systematic approach to processing sales records, identifying important trends, and presenting meaningful information from the available dataset. It helps understand product performance, regional sales distribution, monthly sales variations, and overall revenue generation.

The project is implemented using **Python**, with **Pandas and NumPy** for data processing, manipulation, and numerical calculations. **Matplotlib and Seaborn** are used for data visualization, enabling the representation of sales information through charts and graphs. The integration of these tools provides an effective method for organizing raw sales data, performing calculations, and presenting the results in a clear and understandable format.

One of the main objectives of the project is to **identify sales trends, analyze product performance, compare regional sales, and evaluate revenue patterns**. By performing data cleaning, handling missing values, removing duplicate records, and calculating revenue from quantity and unit price, the project improves the consistency and usefulness of the dataset. Visual representations of monthly sales and product-wise performance help identify patterns that may not be easily understood from raw tabular data.

Overall, the **Sales Data Analysis** project demonstrates the practical application of data analysis techniques and Python programming in business environments. It transforms raw sales records into meaningful information that can assist in understanding customer demand, monitoring business performance, and supporting data-driven decisions. The project highlights how data analysis and visualization can improve the interpretation of sales information and contribute to better business planning.

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1. INTRODUCTION

Sales Data Analysis is an important process used to understand business performance by examining sales records, identifying patterns, and converting raw information into useful insights. In a competitive business environment, organizations generate large amounts of data through orders, products, customers, dates, and regions. When these records are studied systematically, they help an organization understand which products perform well, which regions contribute more revenue, and how sales change over time.

Traditional sales monitoring often depends on manually prepared spreadsheets and reports. When the volume of records increases, manual calculations can take considerable time and may lead to duplicate entries, incorrect totals, inconsistent date formats, and difficulty comparing sales across products or regions. These problems can affect the accuracy of business decisions. A structured sales data analysis process helps reduce such difficulties by organizing the data, checking its quality, performing calculations consistently, and presenting the results in a clear form.

The primary purpose of this project is to analyse sales transaction data and present meaningful information about revenue, product performance, regional performance, and monthly sales trends. The project uses a dataset containing fields such as Order ID, Date, Product, Region, Quantity, and Unit Price. Revenue is calculated by multiplying the quantity sold by the unit price. The calculated values can then be grouped and compared to identify useful patterns in the available records.

The project is implemented using Python, a programming language widely used for data processing and analysis. Pandas is used to load tabular data, inspect columns, clean records, and group information for analysis. NumPy supports numerical operations, while Matplotlib and Seaborn are used to create charts that make the results easier to understand. These tools provide a repeatable workflow in which the same steps can be applied whenever an updated sales dataset becomes available.

The analysis process begins with loading the source file and examining its structure. The dataset is checked for missing values, duplicate rows, inconsistent data types, and invalid numerical values. Date fields are converted into a suitable date format, and quantity and unit-price fields are checked before revenue is calculated. Cleaning is performed carefully so that errors are identified rather than hidden. Any assumptions made during preparation should be documented, especially when the dataset contains incomplete or illustrative records.

After preprocessing, the project performs product-wise, region-wise, and month-wise analysis. Product-wise analysis compares revenue generated by different products and helps identify stronger and weaker contributors. Region-wise analysis compares the sales contribution of each region. Monthly analysis groups transactions by month to show changes in sales over time and to reveal possible increases, decreases, or recurring patterns. Charts and summary tables are used to communicate these findings in a concise and understandable way.

The results of sales data analysis can support planning, inventory control, marketing decisions, and performance monitoring. For example, a product with consistently high sales may require closer stock monitoring, while a region with lower sales may need further investigation. Monthly trends can help a business plan promotions or prepare for periods of higher demand. However, the analysis describes the records provided; it does not automatically prove the reasons behind a trend. Additional information may be required before making operational decisions.

The project also demonstrates the importance of data quality and reproducibility. A chart is only as reliable as the information used to create it, so cleaning and validation are essential stages. The steps are organized so that the analysis can be repeated and reviewed. The sample dataset used for demonstration is illustrative and should not be interpreted as actual confidential sales information from the internship organization. When official data is available, it can be substituted using the same documented workflow.

Overall, the Sales Data Analysis project provides practical experience in data preparation, numerical calculation, exploratory analysis, and visualization. It connects programming concepts with a common business requirement: transforming transaction records into information that can be interpreted and used. The following sections describe

the internship objectives, project scope, organization profile, system design, implementation methodology, testing approach, results, limitations, and possible future enhancements.

1.1 OBJECTIVES OF THE INTERNSHIP

Specific Objectives

The main objective of the Sales Data Analysis project is to examine sales records and transform raw transaction data into useful information for understanding business performance. The project uses Python-based data analysis techniques to study revenue, product performance, regional sales, and monthly trends in a clear and systematic manner.

The objectives of the project are:

- To develop a structured sales data analysis workflow using Python and suitable data analysis libraries.
- To load sales records from a CSV file and inspect the dataset's rows, columns, data types, and overall structure.
- To identify and handle missing values, duplicate records, inconsistent formats, and invalid data where appropriate.
- To prepare date, quantity, and unit-price fields in a consistent format for reliable analysis.
- To calculate sales revenue using the formula: **Revenue = Quantity × Unit Price**.
- To analyse product-wise sales and compare the revenue generated by different products.
- To examine region-wise sales and compare the contribution of each region to overall revenue.
- To group transactions by month and identify changes and patterns in sales over time.
- To use Pandas and NumPy for data manipulation, numerical calculations, grouping, and summary statistics.
- To create clear charts using Matplotlib and Seaborn to communicate the analysis results effectively.
- To present key findings in readable tables and visual summaries that can support business planning.
- To validate calculations and review the results so that errors and unsupported conclusions are avoided.
- To make the analysis repeatable so that updated sales files can be processed using the same workflow.
- To demonstrate practical skills in data cleaning, exploratory data analysis, programming, and visualization.

Expected Outcome

The expected outcome is a clean and organized analysis of the available sales dataset, including revenue calculations, product-wise and region-wise summaries, monthly sales comparisons, and supporting visualizations. The dataset used for demonstration should be treated as illustrative unless it is replaced with verified project data.

1.2 SCOPE OF THE PROJECT

The scope of the project is limited to the analysis of sales transaction records stored in a CSV file. Each record contains the Order ID, Date, Product, Region, Quantity, and Unit Price of a sale. The work covers loading and inspecting the file, cleaning and preparing the data, calculating revenue, and comparing sales by product, region, and month. The results are presented through summary tables and charts.

The following activities are included within the scope of the project:

- Loading and inspection of the sales dataset using Pandas.
- Handling of duplicate records, missing values, inconsistent text formats, and date conversion.
- Calculation of total revenue, total orders, and average order value.
- Product-wise, region-wise, and monthly sales analysis.
- Creation of bar charts and a line chart using Matplotlib.
- Validation of the results by checking that the grouped totals match the total revenue.

Activities such as sales forecasting, profit analysis, customer-wise analysis, interactive dashboards, and loading data directly from a database are not part of the present work. They are listed under Future Work as possible enhancements.

1.3 ORGANIZATION OF THE REPORT

The report is organized into thirteen chapters. Table 1.1 gives a short description of each chapter.

Chapter	Content
1	Introduction, objectives of the internship and scope of the project
2	Profile of Fantasy Solution and the role played during the internship
3	Daily work report of the 30-day internship
4	Block diagram of the analysis workflow and its explanation
5	Existing system, proposed system, capabilities, dataset and requirements
6	Proposed method used for each stage of the analysis
7	Step-by-step algorithms for each module
8	Python source code of the project
9	Sample outputs, charts and result summary tables
10	Analysis, testing, validation, limitations and business implications
11	Conclusion and learning outcomes
12	Possible future enhancements
13	References

Table 1.1: Organization of the report

1.4 KEY TERMS USED IN THE REPORT

Table 1.2 explains the main terms used in this report so that the analysis can be followed easily.

Term	Meaning
CSV file	A plain text file in which each line is one record and the fields are separated by commas. The sales data is stored in this form.
DataFrame	The table-like structure of Pandas in which the sales records are held as rows and columns.
Revenue	The money earned from an order, calculated as Quantity × Unit Price.
Average Order Value	Total revenue divided by the number of orders.
Median	The middle value of a set of numbers. It is used to fill the missing quantity value.
Grouping (Aggregation)	Collecting records that share a product, region or month and adding their revenue together.
Data Cleaning	Correcting or removing duplicate, missing and inconsistent records before analysis.
Exploratory Data Analysis	Studying the data with summaries and charts to find patterns and differences.
Data Visualization	Showing the results as bar charts and line charts so that they are easy to understand.

Table 1.2: Key terms

2. COMPANY PROFILE

About the Company:

Fantasy Solution is an Information Technology (IT) company that focuses on providing software development and digital technology solutions to meet the growing needs of modern businesses. The company specializes in developing web applications, software solutions, and technology-based services that help organizations improve operational efficiency and adopt digital solutions. Its primary objective is to develop reliable, user-friendly, and efficient software applications that address real-world business requirements.

The company follows modern software development practices and focuses on application design, development, database management, testing, and implementation. It emphasizes quality, performance, maintainability, and customer satisfaction throughout the development process.

Fantasy Solution also provides industrial training and internship opportunities for students and fresh graduates. These programs help students connect academic knowledge with practical industry requirements through exposure to real-time projects and commonly used development tools.

Internship Experience:

As part of my internship at Fantasy Solution, I worked on a Sales Data Analysis project. The project focuses on examining sales records and transforming raw transaction data into useful information for understanding business performance. The work includes loading sales records from a CSV file, inspecting the dataset, preparing and cleaning data, calculating revenue, and comparing product-wise, region-wise, and monthly sales.

Python is used for the analysis, with Pandas and NumPy supporting data manipulation, numerical calculations, grouping, and summary statistics. Matplotlib and Seaborn are used to create charts that communicate the results. The project provides practical experience in data cleaning, exploratory data analysis, programming, and data visualization. The findings are intended to support business planning while avoiding unsupported conclusions.

Role During the Internship:

During the internship my work was centred on the Sales Data Analysis project. My main responsibilities were:

- Loading the sales file and inspecting its rows, columns and data types.
- Cleaning the data by handling duplicates, missing values and inconsistent text formats.
- Calculating revenue and the overall sales metrics.
- Carrying out product-wise, region-wise and monthly analysis.
- Creating charts and summary tables to present the findings.
- Checking the results and documenting the work in this report.

Company Details:

- **Company Name:** Fantasy Solution
- **Industry:** Information Technology (IT) Services and Software Development
- **Location:** # 16, Samanth Plazza, Madurai Main Road, Melapudur, Trichy-1, Tamil Nadu, India
- **Development Technologies:** Java, Spring Boot, HTML, CSS, JavaScript, Bootstrap, MySQL, REST APIs
- **Major Services:** Software Development, Web Application Development, Database Solutions, Mobile Application Development, Software Testing, and Technical Training
- **Internship Training:** Web Application Development, Database Management, REST API Development, Software Testing, and Real-Time Projects
- **Development Approach:** User-friendly, scalable, secure, reliable, and industry-oriented software solutions
- **Project Completed During Internship:** Sales Data Analysis (Python-based Data Analysis)

Services Offered

- Software Development
- Web Application Development
- Website Design and Development
- Mobile Application Development
- Java Full Stack Development
- Database Design and Management
- REST API Development
- Digital Marketing and SEO
- Domain Registration and Web Hosting
- Technical Internship and Industrial Training
- Software Maintenance and Technical Support

Technologies Used by the Company

- **Programming Languages:** Java, JavaScript, Python
- **Frontend:** HTML5, CSS3, JavaScript

- **Backend:** Java Spring Boot
- **Database:** MySQL
- **Data Analysis Libraries:** Pandas, NumPy
- **Visualization Libraries:** Matplotlib, Seaborn
- **Development Tools:** Eclipse, IntelliJ IDEA, VS Code
- **Build Tool:** Apache Maven
- **Version Control:** Git and GitHub
- **API Technology:** RESTful Web Services

3. DAILY WORK REPORT

Day 1: As a beginner in data science, I familiarized myself with the foundational concepts of statistics, probability, and data visualization. I learned about descriptive statistics and explored tools like Python and R for data analysis.

Day 2: On the second day, I delved into data preprocessing, understanding techniques for cleaning and transforming raw data. I learned about handling missing values, outlier detection, and feature scaling to ensure data quality and reliability.

Day 3: During day three, I explored data exploration and visualization, utilizing libraries like Matplotlib and Seaborn to create insightful visual representations of data. I learned how to uncover patterns, trends, and relationships within datasets.

Day 4: On the fourth day, I ventured into the world of machine learning, understanding the basic concepts of supervised and unsupervised learning. I explored algorithms like linear regression, decision trees, and clustering.

Day 5: During the fifth day, I focused on evaluating and validating machine learning models. I learned about metrics like accuracy, precision, and recall, and techniques like cross-validation and train-test splitting.

Day 6: On the sixth day, I delved into feature engineering, understanding how to extract meaningful features from raw data. I explored techniques like one-hot encoding, feature scaling, and dimensionality reduction.

Day 7: During day seven, I explored the concept of model selection and optimization. I learned about hyper parameter tuning, grid search, and techniques like regularization to improve model performance.

Day 8: On the eighth day, I ventured into the world of classification algorithms, understanding techniques like logistic regression, support vector machines, and random forests for predicting categorical outcomes.

Day 9: During the ninth day, I focused on regression algorithms, learning about techniques like linear regression, polynomial regression, and gradient boosting for predicting continuous variables.

Day 10: On the tenth day, I delved into clustering algorithms, understanding techniques like k-means, hierarchical clustering, and DBSCAN for grouping similar data points together.

Day 11: During day eleven, I explored the concept of dimensionality reduction, understanding techniques like principal component analysis (PCA) and t-SNE for visualizing and reducing high-dimensional data.

Day 12: On the twelfth day, I delved into natural language processing (NLP), learning about techniques for text preprocessing, sentiment analysis, and text classification using libraries like NLTK and spaCy.

Day 13: During the thirteenth day, I focused on time series analysis, understanding techniques like ARIMA, seasonal decomposition, and forecasting models for analyzing and predicting sequential data.

Day 14: On the fourteenth day, I ventured into the world of recommendation systems, understanding collaborative filtering and content-based approaches for providing personalized recommendations.

Day 15: During the fifteenth day, I explored the concept of ensemble learning, understanding techniques like bagging, boosting, and stacking for combining multiple models to improve overall performance.

Day 16: On the sixteenth day, I delved into the world of deep learning, understanding concepts like artificial neural networks, convolutional neural networks (CNN), and recurrent neural networks (RNN).

Day 17: During day seventeen, I focused on understanding the fundamentals of neural network architecture, learning about layers, activation functions, and optimization algorithms like gradient descent.

Day 18: On the eighteenth day, I explored the concept of transfer learning in deep learning, understanding how pre-trained models can be used for various tasks and domains.

Day 19: During the nineteenth day, I delved into the world of computer vision, understanding techniques like image classification, object detection, and image segmentation using deep learning models.

Day 20: On the twentieth day, I focused on understanding the basics of natural language processing (NLP) with deep learning, exploring techniques like word embeddings and recurrent neural networks (RNN).

Day 21: During day twenty-one, I ventured into the world of data visualization, learning about libraries like Tableau and Plotly for creating interactive and visually appealing visualizations.

Day 22: On the twenty-second day, I delved into the world of big data and distributed computing, understanding concepts like Hadoop, Spark, and MapReduce for handling large datasets.

Day 23: During the twenty-third day, I focused on understanding the basics of SQL and databases, learning how to write queries, create tables, and perform data manipulation using SQL.

Day 24: On the twenty-fourth day, I explored the integration of machine learning models with real-time data pipelines. I learned how tools like Apache Kafka and Spark Streaming can be used to process and analyze data streams in real-time, enabling dynamic and responsive machine learning applications.

Day 25: During day twenty-five, I focused on model deployment techniques, exploring tools such as Flask and FastAPI to create APIs for machine learning models. I also learned about using Docker for containerization and how to deploy models on cloud platforms like AWS and Google Cloud.

Day 26: On the twenty-sixth day, I concentrated on monitoring and maintaining deployed machine learning models. I studied how to track performance over time using logging and monitoring tools, and how to implement retraining strategies when the model's accuracy begins to degrade due to concept drift or changing data patterns.

Day 27: This day was dedicated to review and reflection. I revisited key concepts learned throughout the internship, from data preprocessing to deployment. I also prepared a summary presentation and shared insights gained, challenges faced, and how the internship contributed to my growth in data mining, machine learning, and real-world application development.

Day 28: On the twenty-eighth day, I ventured into the world of data pipelines and automation, understanding how to streamline data workflows using tools like Apache Airflow and cron jobs.

Day 29: During the twenty-ninth day, I focused on understanding the basics of cloud computing and storage, exploring platforms like AWS and GCP for scalable and secure data solutions.

Day 30: On the final day, I reviewed the complete Sales Data Analysis workflow, covering data loading, cleaning, revenue calculation, product-wise, region-wise and monthly analysis, and chart generation. I checked the results and compiled the final internship report.

3.1 Summary of the Internship Period

Table 3.1 summarizes the topics covered during the 30 days of the internship.

Days	Topics Covered
Day 1 – Day 7	Statistics, data preprocessing, data visualization, basics of machine learning, model evaluation, feature engineering, and model selection and optimization.
Day 8 – Day 15	Classification, regression and clustering algorithms, dimensionality reduction, natural language processing, time series analysis, recommendation systems and ensemble learning.
Day 16 – Day 23	Deep learning, neural network architecture, transfer learning, computer vision, NLP with deep learning, visualization tools, big data and SQL.
Day 24 – Day 30	Real-time data pipelines, model deployment, monitoring of deployed models, review and reflection, data pipelines and automation, cloud computing, and review of the Sales Data Analysis workflow.

Table 3.1: Summary of the internship period

4. BLOCK DIAGRAM

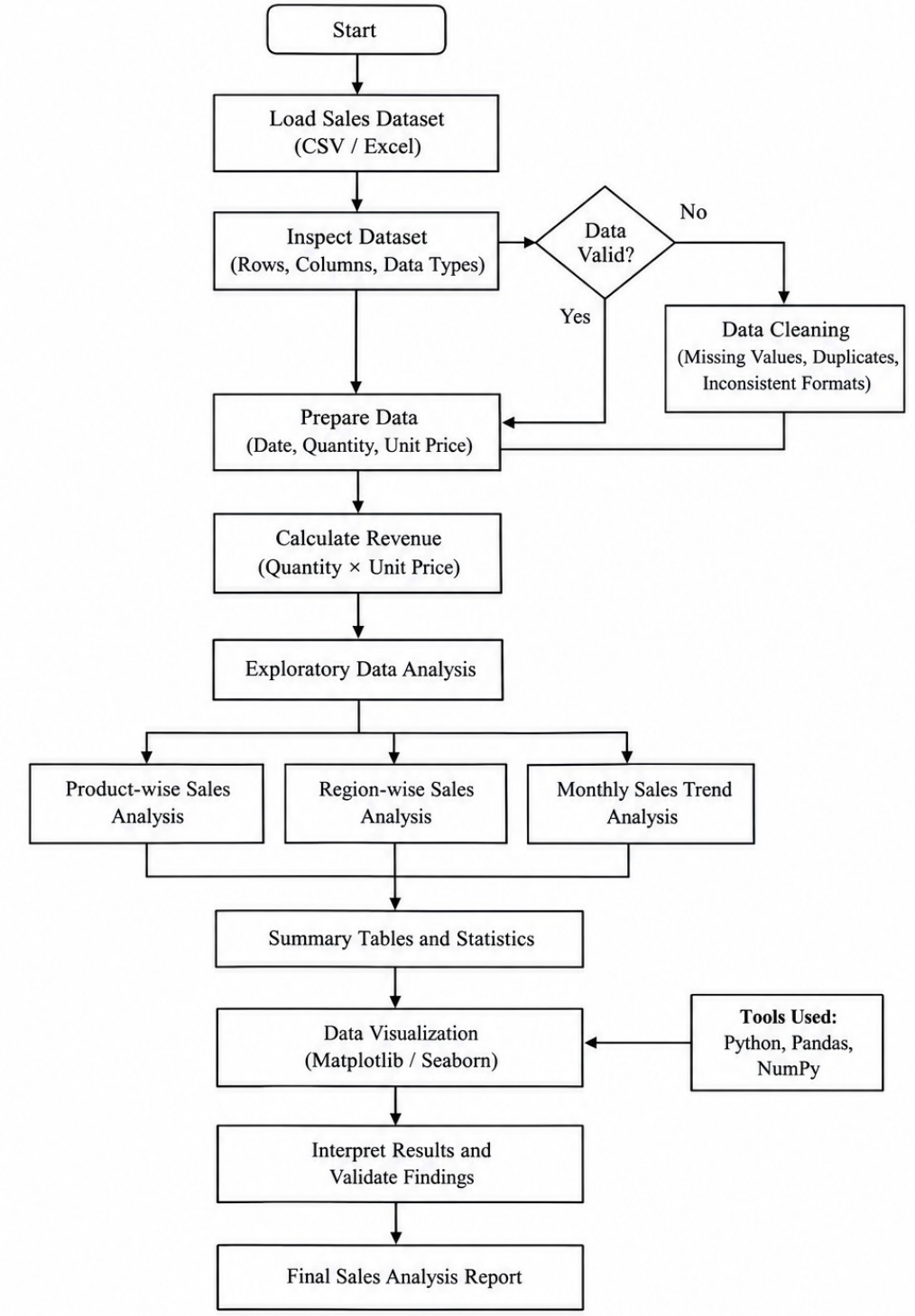


Figure 4.1: Block diagram of the Sales Data Analysis workflow

4.1 EXPLANATION OF THE BLOCK DIAGRAM

The block diagram shows the complete workflow of the Sales Data Analysis project, from loading the raw sales file to preparing the final report. The workflow is divided into stages, and the output of each stage becomes the input of the next stage. Table 4.1 explains the purpose of each stage.

Stage	Description
Load Sales Dataset	The sales file (CSV or Excel) is read into a Pandas DataFrame.
Inspect Dataset	The rows, columns and data types are examined, and the data is checked for missing values and duplicate records.
Data Valid?	If the data is valid it moves directly to preparation; otherwise it is sent for cleaning first.
Data Cleaning	Missing values, duplicates and inconsistent formats are corrected so that the records can be trusted.
Prepare Data	The Date, Quantity and Unit Price fields are converted into suitable date and numeric formats.
Calculate Revenue	Revenue is calculated for each order using Quantity × Unit Price.
Exploratory Data Analysis	The prepared data is explored from three viewpoints: product, region and month.
Summary Tables and Statistics	Product-wise, region-wise and monthly totals are collected into summary tables.
Data Visualization	Bar charts and a line chart are created with Matplotlib and Seaborn.
Interpret Results and Validate Findings	The results are reviewed, the totals are cross-checked, and only supported conclusions are drawn.
Final Sales Analysis Report	The tables, charts and findings are compiled into the final report.

Table 4.1: Stages of the Sales Data Analysis workflow

The decision block “Data Valid?” is an important part of the workflow because it ensures that no record is analysed before its quality has been checked. Data that passes the check goes straight to preparation, while data that fails is cleaned first and then rejoins the main flow.

5. SALES DATA ANALYSIS

5.1 Existing System

In many organizations, sales are still monitored through manually prepared spreadsheets and periodic reports. Transaction records containing the order ID, date, product, region, quantity and unit price are entered and totalled by hand, and separate summaries for products, regions and months are prepared again whenever new records arrive. When the number of records grows, this approach becomes slow and error-prone. The existing approach suffers from several limitations:

- **Manual calculation** – Revenue for each order and the overall totals are calculated manually, increasing the chances of incorrect totals.
- **Duplicate and incomplete records** – Repeated orders, missing quantities and blank region entries often remain unnoticed in the sheet.
- **Inconsistent formats** – Dates, product names and region names are entered in different styles, such as extra spaces or lower-case letters, which makes grouping unreliable.
- **Time-consuming reporting** – Product-wise, region-wise and monthly summaries must be rebuilt by hand each time the data changes.
- **Difficulty in comparison** – Comparing the contribution of different products and regions to the overall revenue is difficult using raw records.
- **Limited visualization** – Charts are prepared separately and are not easily repeated for updated files.
- **No repeatability** – The manual steps are not documented as a fixed workflow, so results may differ from one preparation to another.

These drawbacks make the manual method less suitable for growing volumes of sales records. Therefore, there is a need for an automated and repeatable analysis process that cleans the data, calculates revenue consistently and presents the results clearly.

5.2 Proposed System

The proposed system is a Python-based Sales Data Analysis workflow that converts raw transaction data into useful information. It loads the sales file, inspects its structure, cleans the records, calculates revenue and compares sales by product, region and month. The results are presented through summary tables and charts. The system uses Python

with Pandas and NumPy for data processing and Matplotlib and Seaborn for visualization. Key aspects of the proposed system include:

- **Automated data loading** – Sales records are loaded from a CSV file and inspected for rows, columns, data types and non-null values.
- **Data cleaning** – Duplicate orders are removed, dates are converted to a consistent format, missing quantities are filled, records without product or region are dropped, and text fields are standardized.
- **Revenue calculation** – Revenue is calculated for every order using Quantity × Unit Price, and overall metrics are computed.
- **Product-wise and region-wise analysis** – Revenue is grouped and sorted to identify the top product and the top region.
- **Monthly trend analysis** – Revenue is grouped by month to show increases, decreases and patterns over time.
- **Data visualization** – Bar charts and a line chart are created and saved for use in the report.
- **Repeatable workflow** – The same script can be run on an updated sales file without changing the logic.

By automating these steps, the proposed system reduces manual effort, removes calculation errors caused by duplicate and blank records, and provides reliable results that can support business planning.

5.3 Capabilities

The Sales Data Analysis system offers the following key capabilities.

- **Data Loading and Inspection** – Reads the sales file and displays its first rows, column names, data types and non-null counts.
- **Data Cleaning and Preparation** – Handles duplicates, missing values, date conversion and inconsistent text formats.
- **Revenue Calculation** – Calculates revenue for each order and computes total revenue, total orders and average order value.
- **Product-Wise Analysis** – Compares the revenue generated by each product and identifies the top-selling product.
- **Region-Wise Analysis** – Compares the revenue of each region and identifies the best-performing region.
- **Monthly Sales Analysis** – Shows how revenue changes from month to month and identifies the best month.
- **Data Visualization** – Produces bar charts and a line chart that present the results clearly.
- **Validation** – Product-wise, region-wise and monthly totals are checked against the total revenue to confirm that no record is lost or counted twice.

5.4 Dataset Description

The analysis uses a sales transaction file named sales_data.csv. When the file is loaded, it contains 61 rows and 6 columns, recording sales made from January to June 2025. The fields of the dataset are described in Table 5.1.

Field	Description	Data Type
Order_ID	Unique identifier of each sales order (for example, ORD1001)	Text
Date	Date on which the order was placed	Text (converted to date)
Product	Name of the product sold	Text
Region	Sales region of the order: North, East, West or South	Text
Quantity	Number of units sold in the order	Decimal number
Unit_Price	Price of one unit of the product, in INR	Integer

Table 5.1: Fields of the sales dataset

The raw file also contains some imperfections that are handled during cleaning: one duplicate order, one missing quantity, one missing region, and region names written with extra spaces or in lower-case letters. After cleaning, 59 valid orders remain. The products sold are Smartphone, Tablet, Laptop, Smartwatch and Headphones. The data used in this report is illustrative sample data.

5.5 System Requirements

The Sales Data Analysis script is lightweight and does not need special hardware. The software and hardware requirements for running it are given in Table 5.2.

Requirement	Details
Programming Language	Python
Libraries	Pandas, NumPy, Matplotlib and Seaborn
Input Data	Sales records in CSV or Excel format
Development Environment	VS Code or Jupyter Notebook
Operating System	Windows, Linux or macOS
Hardware	A standard computer or laptop capable of running Python

Table 5.2: Software and hardware requirements

5.6 Comparison of Existing and Proposed Systems

Table 5.3 compares the manual spreadsheet method described in Section 5.1 with the proposed Python-based workflow described in Section 5.2.

Aspect	Existing System	Proposed System
Calculation	Revenue and totals are calculated manually	Revenue is calculated automatically as Quantity × Unit Price
Data quality	Duplicate and blank records often go unnoticed	Duplicates are removed and missing values are handled in a fixed step
Formats	Dates and names are entered in different styles	Dates and text fields are standardized by the script
Reporting	Summaries are rebuilt by hand whenever data changes	Summaries are produced by running the script again
Comparison	Difficult to compare products and regions from raw records	Grouped totals are sorted and compared directly
Charts	Prepared separately and not easily repeated	Created and saved by the same script
Repeatability	Results may differ between preparations	The same steps give the same results every time

Table 5.3: Existing system versus proposed system

5.7 Advantages of the Proposed System

- **Saves time:** Cleaning, calculation and charting are done in a single run of the script.
- **Reduces errors:** Duplicate and blank records no longer distort the totals.
- **Consistent results:** Revenue is always calculated with the same formula.
- **Clear presentation:** Tables and charts make comparisons easy to understand.
- **Easy to reuse:** The same workflow can be applied to a new sales file without changing the logic.
- **Easy to extend:** Further analysis can be added to the script as new requirements arise.

6. PROPOSED METHOD

The proposed method divides the Sales Data Analysis project into clear stages. Each stage takes the output of the previous stage as its input, so that the quality of the data is checked before any calculation is made and the results are validated before any conclusion is drawn. The stages are described below, and a summary of their inputs and outputs is given in Section 6.10.

6.1 Dataset Loading and Inspection

- Loads sales records from a CSV or Excel file using Python.
- Examines dataset rows, columns, data types, and overall structure.
- Identifies important fields such as date, product, quantity, unit price, and region.

- Checks the dataset for missing values, duplicate records, and inconsistent formats.
- Confirms that the data is suitable for further analysis.

6.2 Data Cleaning and Preparation

- Handles missing values and removes duplicate records where appropriate.
- Converts date fields into a consistent date format.
- Ensures quantity and unit-price fields use valid numeric values.
- Standardizes product names, region names, and other categorical fields.
- Prepares a clean dataset for accurate calculations and comparison.

6.3 Revenue Calculation

- Calculates sales revenue for each transaction using Quantity × Unit Price.
- Adds a revenue field to the prepared dataset.
- Checks calculations and reviews unusual or invalid values.
- Aggregates revenue for overall sales summaries.
- Produces consistent values for product, region, and monthly analysis.

6.4 Product-wise Sales Analysis

- Groups sales records by product name.
- Calculates total quantity sold and revenue generated by each product.
- Compares product performance to identify high- and low-performing products.
- Presents product-wise results in a summary table.
- Uses charts to make product comparisons easier to understand.

6.5 Region-wise Sales Analysis

- Groups sales transactions according to region.
- Calculates total revenue and sales contribution for each region.
- Compares regional performance to identify differences in sales.
- Organizes regional results in clear summary tables.
- Uses visualizations to communicate region-wise sales patterns.

6.6 Monthly Sales Trend Analysis

- Converts transaction dates into a consistent date format.
- Groups sales records by month and calculates monthly revenue.
- Compares monthly totals to identify increases, decreases, and patterns.
- Displays monthly sales trends using line or bar charts.
- Uses the observed trends to support careful business interpretation.

6.7 Data Visualization and Summary

- Uses Pandas and NumPy to organize data and calculate summary statistics.
- Creates bar charts to compare revenue across products and regions.
- Uses line charts to display monthly sales trends.
- Labels charts clearly and selects suitable titles and scales.
- Presents results in readable tables and visual summaries.

6.8 Result Interpretation and Report Generation

- Reviews product-wise, region-wise, and monthly sales results.
- Checks calculations and avoids conclusions not supported by the data.

- Identifies useful patterns and differences in sales performance.
- Summarizes the main findings for business planning.
- Prepares the final Sales Data Analysis report with tables, charts, and conclusions.

6.9 Tools and Technologies Used

- **Programming Language:** Python.
- **Data Manipulation:** Pandas and NumPy.
- **Data Visualization:** Matplotlib and Seaborn.
- **Input Data:** CSV or Excel sales records.
- **Output:** Summary tables, charts, and a final analysis report.

6.10 Workflow Summary

Table 6.1 summarizes the input and output of each stage of the proposed method.

Stage	Input	Output
Loading	sales_data.csv	DataFrame of 61 rows and 6 columns
Cleaning	Raw DataFrame	Clean dataset of 59 valid orders
Revenue calculation	Clean dataset	Revenue column and overall metrics
Analysis	Clean dataset with revenue	Product-wise, region-wise and monthly totals
Visualization	Grouped totals	Three chart images (PNG files)
Validation	Grouped totals	Confirmation that each total equals INR 2,923,500

Table 6.1: Input and output of each stage

7. ALGORITHM

The algorithms below describe the steps carried out by each module of the script. They are written in the order in which the modules run, and the output of each module is passed to the next one.

7.1 Data Loading and Inspection – Algorithm

1. Start the program and import the required libraries (Pandas and Matplotlib).
2. Read the sales file (sales_data.csv) into a Pandas DataFrame.
3. Display the first rows of the DataFrame.
4. Display the column names, data types and non-null counts.
5. Check the data for missing values and duplicate records.
6. Pass the loaded data to the cleaning step.

7.2 Data Cleaning and Preprocessing – Algorithm

1. Remove duplicate orders from the DataFrame.
2. Convert the Date column into a consistent date format.
3. Calculate the median of the Quantity column.
4. Fill the missing quantity values with the median.
5. Remove the records in which Product or Region is missing.
6. Strip extra spaces and convert the Region names to title case.
7. Strip extra spaces and convert the Product names to title case.
8. Pass the cleaned dataset to the next step.

7.3 Revenue Calculation – Algorithm

1. For every order, multiply Quantity by Unit Price to create the Revenue column.

2. Add all revenue values to obtain the total revenue.
3. Count the unique Order IDs to obtain the total number of orders.
4. Calculate the mean of the Revenue column to obtain the average order value.
5. Display the total revenue, total orders and average order value.

7.4 Product-Wise Analysis – Algorithm

1. Group the records by Product.
2. Add the revenue of each product.
3. Sort the products in descending order of revenue.
4. Display the product-wise revenue.
5. Identify the product with the highest revenue as the top product.

7.5 Region-Wise Analysis – Algorithm

1. Group the records by Region.
2. Add the revenue of each region.
3. Sort the regions in descending order of revenue.
4. Display the region-wise revenue.
5. Identify the region with the highest revenue as the top region.

7.6 Monthly Sales Analysis – Algorithm

1. Extract the month from the Date column.
2. Group the records by month.
3. Add the revenue of each month.
4. Display the monthly revenue.
5. Identify the month with the highest revenue as the best month.

7.7 Data Visualization – Algorithm

1. Create a bar chart of revenue by product with a title and axis labels, and save it as product_revenue.png.
2. Create a bar chart of revenue by region with a title and axis labels, and save it as region_revenue.png.
3. Convert the month index to text and create a line chart of the monthly sales trend with markers.
4. Save the line chart as monthly_trend.png.
5. End the program.

7.8 Overall Program – Algorithm

1. Start the program and load the sales file (sales_data.csv) into a DataFrame.
2. Inspect the data for missing values and duplicate records.
3. Clean the data: remove duplicates, convert dates, fill missing quantities, drop records without product or region, and standardize text.
4. Calculate the Revenue column and the overall metrics.
5. Group the revenue by product, by region and by month.
6. Identify the top product, the top region and the best month.
7. Create and save the three charts.
8. Check that the product-wise, region-wise and monthly totals are equal to the total revenue.
9. Display the results and end the program.

8. CODE IMPLEMENTATION

The complete source code of the Sales Data Analysis script (sales_analysis.py) is given below, divided into modules.

8.1 Importing Libraries and Loading Data

Imports the required libraries and loads the raw sales file into a DataFrame for inspection.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("sales_data.csv")
print(df.head())
print(df.info())
```

8.2 Data Cleaning and Preprocessing

Removes duplicate orders, fixes data types, fills missing quantities and standardises text fields.

```
df = df.drop_duplicates()
df["Date"] = pd.to_datetime(df["Date"])
df["Quantity"] = df["Quantity"].fillna(
    df["Quantity"].median())
df = df.dropna(subset=["Product", "Region"])
df["Region"] = df["Region"].str.strip().str.title()
df["Product"] = df["Product"].str.strip().str.title()
```

8.3 Revenue Calculation

Calculates revenue for every order and computes the overall sales metrics.

```
df["Revenue"] = df["Quantity"] * df["Unit_Price"]
total_revenue = df["Revenue"].sum()
total_orders = df["Order_ID"].nunique()
average_order_value = df["Revenue"].mean()

print("Total Revenue:", total_revenue)
print("Total Orders:", total_orders)
print("Average Order Value:", average_order_value)
```

8.4 Product-Wise Analysis

Groups the sales by product and identifies the best-performing product.

```
product_sales = (
    df.groupby("Product")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)
print(product_sales)
print("Top Product:", product_sales.idxmax())
```

8.5 Region-Wise Analysis

Groups the sales by region and identifies the best-performing region.

```
region_sales = (
    df.groupby("Region")["Revenue"]
    .sum()
    .sort_values(ascending=False)
)
print(region_sales)
print("Top Region:", region_sales.idxmax())
```

8.6 Monthly Sales Analysis

Aggregates revenue month by month to study the sales trend.

```
df["Month"] = df["Date"].dt.to_period("M")
monthly_sales = df.groupby("Month")["Revenue"].sum()
print(monthly_sales)
print("Best Month:", monthly_sales.idxmax())
```

8.7 Data Visualization

Creates and saves the bar charts and line chart used in the report.

```
product_sales.plot(kind="bar", color="steelblue")
plt.title("Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Revenue (INR)")
plt.tight_layout()
plt.savefig("product_revenue.png")
plt.clf()

region_sales.plot(kind="bar", color="seagreen")
plt.title("Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue (INR)")
plt.tight_layout()
plt.savefig("region_revenue.png")
plt.clf()

monthly_sales.index = monthly_sales.index.astype(str)
monthly_sales.plot(kind="line", marker="o", color="darkorange")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Revenue (INR)")
plt.tight_layout()
plt.savefig("monthly_trend.png")
```

8.8 Input Data Format

The script expects the sales file to contain the six columns shown below. The first five records of the sample file are given in Table 8.1.

Order_ID	Date	Product	Region	Quantity	Unit_Price
ORD1001	2025-01-02	Laptop	North	3	55000
ORD1002	2025-01-09	Smartwatch	East	4	7000
ORD1003	2025-01-15	Smartphone	North	5	18000
ORD1004	2025-01-16	Tablet	North	2	22000
ORD1005	2025-01-17	Smartphone	East	5	18000

Table 8.1: First five records of the sample sales file

9. SAMPLE OUTPUT

9.1 Data Loading Output

Displays the first rows and the structure of the sales file after it is loaded into a Pandas DataFrame. It confirms the column names, data types and the number of non-null values in each column.

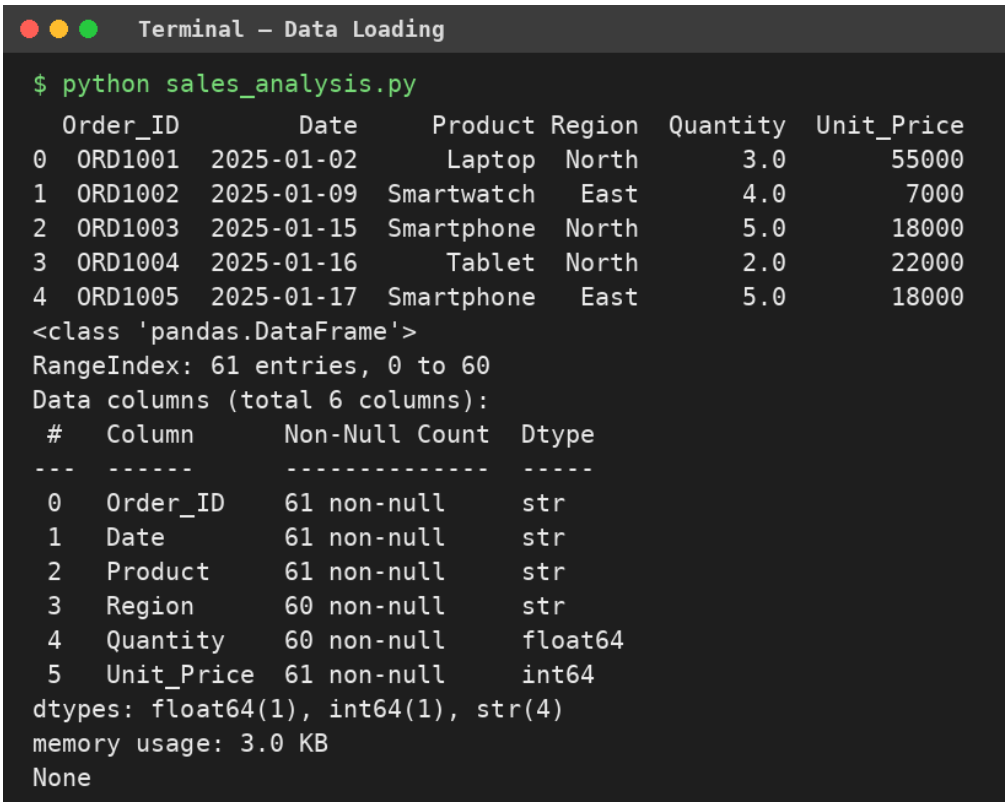


Figure 9.1: Output of Module 8.1

The output shows that the file has 61 entries and 6 columns. The Region and Quantity columns have 60 non-null values each, which shows that one region and one quantity are missing in the raw data. These gaps are handled in the cleaning step.

9.2 Revenue Metrics Output

Displays the overall sales metrics calculated after the data is cleaned. The system reports the total revenue, the total number of valid orders and the average order value.

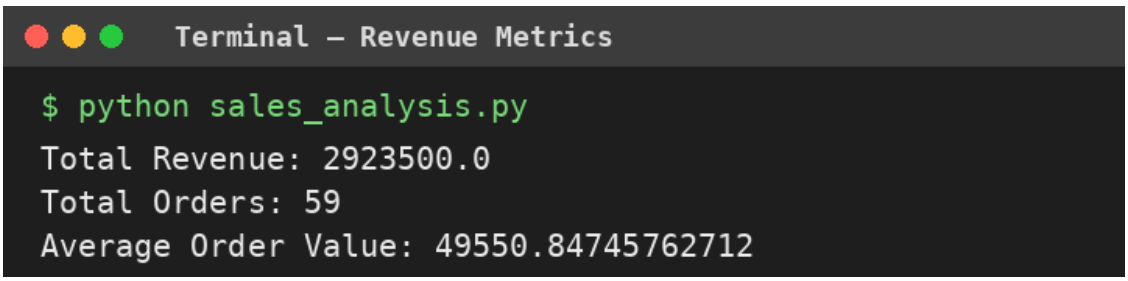


Figure 9.2: Output of Module 8.3

After cleaning, 59 valid orders remain out of the 61 raw rows. The total revenue is INR 2,923,500, and dividing it by the 59 orders gives an average order value of about INR 49,551. These three values are the headline figures used in the rest of the analysis.

9.3 Product-Wise Analysis Output

Displays the revenue earned by each product in descending order and identifies the top-selling product. The bar chart shows the same result visually.

```
Terminal – Product-Wise Analysis
$ python sales_analysis.py
Product
Smartphone    1548000.0
Tablet         594000.0
Laptop         440000.0
Smartwatch     189000.0
Headphones     152500.0
Name: Revenue, dtype: float64
Top Product: Smartphone
```

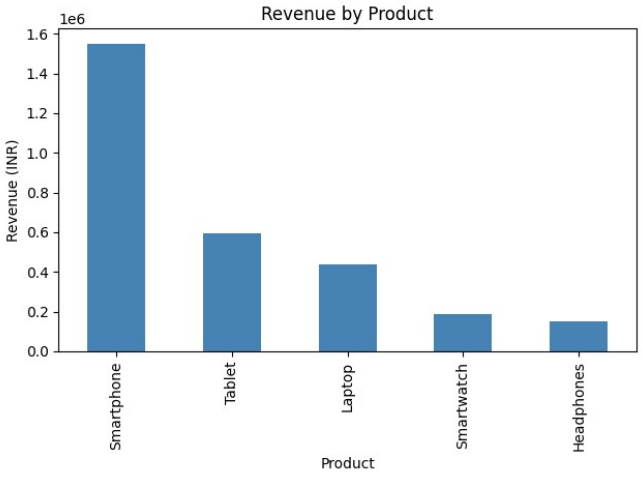


Figure 9.3: Product-wise revenue (Module 8.4 and 8.7)

9.4 Region-Wise Analysis Output

Displays the total revenue of each region and identifies the best-performing region. Administrators can use this output to compare the four sales regions.

```
Terminal – Region-Wise Analysis
$ python sales_analysis.py
Region
North    1121000.0
East      845500.0
West      664000.0
South     293000.0
Name: Revenue, dtype: float64
Top Region: North
```

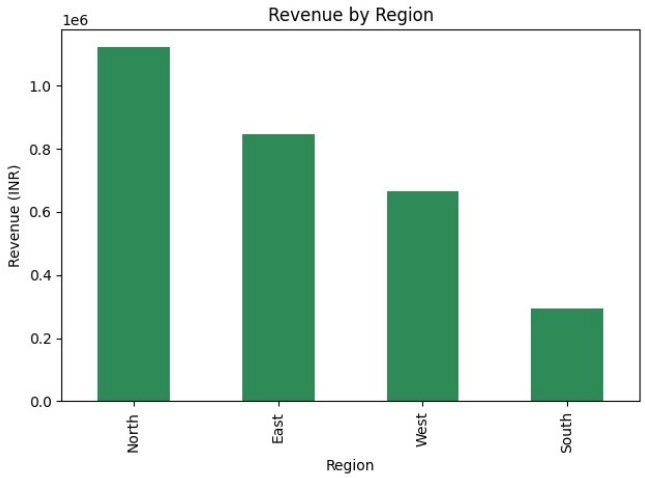


Figure 9.4: Region-wise revenue (Module 8.5 and 8.7)

9.5 Monthly Sales Analysis Output

Displays the revenue for every month from January to June 2025 and identifies the best month. The line chart shows how sales rise and fall over time.

```
Terminal — Monthly Sales Analysis
$ python sales_analysis.py
Month
2025-01    583500.0
2025-02    545000.0
2025-03    368500.0
2025-04    224500.0
2025-05    487000.0
2025-06    715000.0
Freq: M, Name: Revenue, dtype: float64
Best Month: 2025-06
```

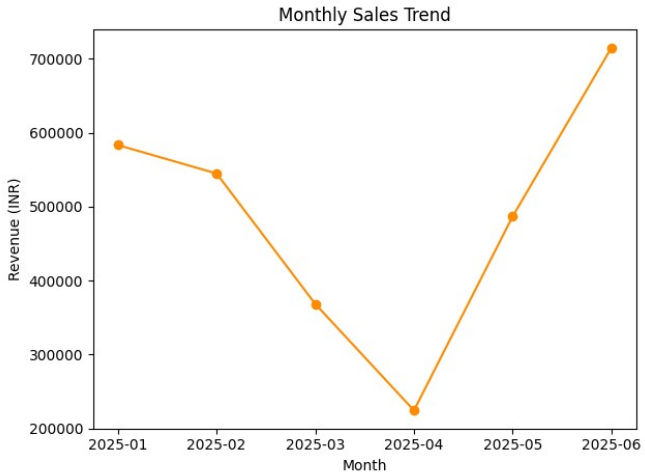


Figure 9.5: Monthly sales trend (Module 8.6 and 8.7)

9.6 Summary of Results

The outputs of the previous sections are consolidated in the tables below. Each share is the revenue of the product, region or month divided by the total revenue of INR 2,923,500.

Product	Revenue (INR)	Share of Total
Smartphone	1,548,000	53.0%
Tablet	594,000	20.3%
Laptop	440,000	15.1%
Smartwatch	189,000	6.5%
Headphones	152,500	5.2%
Total	**2,923,500**	**100.0%**

Table 9.1: Product-wise revenue

Region	Revenue (INR)	Share of Total
North	1,121,000	38.3%
East	845,500	28.9%
West	664,000	22.7%
South	293,000	10.0%
Total	**2,923,500**	**100.0%**

Table 9.2: Region-wise revenue

Month	Revenue (INR)	Share of Total
2025-01 (January)	583,500	20.0%
2025-02 (February)	545,000	18.6%
2025-03 (March)	368,500	12.6%
2025-04 (April)	224,500	7.7%
2025-05 (May)	487,000	16.7%
2025-06 (June)	715,000	24.5%
Total	**2,923,500**	**100.0%**

Table 9.3: Monthly revenue

9.7 Interpretation of the Charts

Revenue by Product (Figure 9.3): The bar for Smartphone is far taller than the others, showing that it earns more than half of the total revenue. Tablet and Laptop form a middle group, while Smartwatch and Headphones have the shortest bars.

Revenue by Region (Figure 9.4): The bars fall steadily from North to East, West and South. The North region earns more than three times as much revenue as the South region.

Monthly Sales Trend (Figure 9.5): The line falls from January to April, where it reaches its lowest point, and then rises sharply in May and June. June is the highest point of the six months.

10. ANALYSIS

Testing of the Sales Data Analysis script was carried out module by module to validate both the correctness of the calculations and the overall behaviour of the program. Functional analysis confirmed that data loading, cleaning, revenue calculation, product-wise, region-wise and monthly analysis and chart generation worked as expected, including the edge cases present in the file such as a duplicate order, a missing quantity, a missing region and region names with extra spaces or lower-case letters. Of the 61 raw rows, 59 valid orders remained after cleaning.

The overall result shows a total revenue of INR 2,923,500 from 59 orders, with an average order value of INR 49,551. The sum of the product-wise totals, the region-wise totals and the monthly totals was checked and found to be equal to the total revenue, which confirms that no record was lost or counted twice during grouping.

10.1 Functional Testing

The test cases used to check the cleaning and calculation steps are listed in Table 10.1.

No.	Test Case	Expected Behaviour	Result
1	Duplicate order present in the file	The repeated order is removed	Passed
2	Date values stored as text	Dates are converted to a date format	Passed
3	Missing Quantity value	The value is filled with the column median	Passed
4	Missing Region value	The record is dropped from the dataset	Passed
5	Region names with extra spaces or lower-case letters	Names are stripped and converted to title case	Passed
6	Revenue calculation	Revenue equals Quantity × Unit Price for each order	Passed
7	Grouped totals compared with total revenue	Product, region and monthly totals each equal the total	Passed

Table 10.1: Functional test cases

10.2 Validation of Results

To confirm that no record was lost or counted twice, the grouped totals were added up and compared with the total revenue. All three groupings give the same value, as shown in Table 10.2.

Grouping	Sum of Group Totals (INR)	Matches Total?
Product-wise	2,923,500	Yes
Region-wise	2,923,500	Yes
Month-wise	2,923,500	Yes
Total revenue	**2,923,500**	—

Table 10.2: Reconciliation of grouped totals with total revenue

10.3 Result Interpretation

Product analysis showed that Smartphone is the top product with INR 1,548,000 (53.0% of total revenue), followed by Tablet and Laptop. Headphones contributes the least (5.2%) because of its low unit price, even though it is sold in good numbers.

Region analysis showed that the North region leads with 38.3% of the revenue, followed by East (28.9%) and West (22.7%). The South region contributes only 10.0%, which indicates the greatest scope for improvement.

Monthly analysis showed that sales were strong in January (INR 583,500) and February (INR 545,000), fell to the lowest value in April (INR 224,500) and then recovered to the highest value in June (INR 715,000). This pattern helps in planning stock and promotional campaigns for the weaker months.

Performance observations showed that the complete script runs within a few seconds on the sample file, because Pandas performs grouping and aggregation efficiently. The same code can be applied to a larger file without any change to the logic.

Comparative analysis against the existing manual spreadsheet method highlighted clear improvements. Cleaning, calculation and charting were reduced from many manual steps to a single run of the script, calculation errors caused by duplicate and blank records were removed, and the results are repeatable every time the script is run on a new file.

Overall, the analysis confirms that the project meets its objectives of cleaning raw sales data, calculating reliable metrics and presenting the findings clearly through tables and charts.

10.4 Limitations

The following limitations should be kept in mind when reading the results:

- The figures in this report are based on illustrative sample data and should be replaced with the actual company data before final submission.
- The dataset covers only six months (January to June 2025), so seasonal patterns over a full year cannot be confirmed.
- The analysis describes what happened in the records; it does not prove the reasons behind a trend, such as the fall in April.
- Only revenue is analysed. Costs and profit are not part of the dataset.

10.5 Business Implications

The results can support planning decisions, provided the findings are first confirmed with actual company data:

- **Stock monitoring:** Smartphone contributes more than half of the revenue, so its stock levels may need closer monitoring.
- **Regional focus:** The South region contributes the smallest share, so it may deserve further investigation.
- **Promotion planning:** Sales were lowest in April and highest in June, which can help in planning promotions and preparing for periods of higher demand.

11. CONCLUSION

The Sales Data Analysis project successfully demonstrates how Python can be used to convert raw sales transaction data into useful information for understanding business performance. By replacing manual spreadsheet work with a repeatable script, the project reduces manual effort, removes errors caused by duplicate and blank records, and presents the results through clear tables and charts.

The data cleaning stage handled a duplicate order, a missing quantity, a missing region and region names with extra spaces or lower-case letters, so that 59 valid orders remained from the 61 raw rows. The analysis showed a total revenue of INR 2,923,500 with an average order value of INR 49,551. Smartphone was the top product, North was the

leading region, and monthly sales were lowest in April and highest in June. The product-wise, region-wise and monthly totals were each checked against the total revenue to confirm the accuracy of the results.

The internship at Fantasy Solution also provided practical experience in Python programming, data cleaning, exploratory data analysis and data visualization using Pandas, NumPy, Matplotlib and Seaborn. The figures in this report are based on illustrative sample data, and the same workflow can be applied to actual company data when it is available.

Learning Outcomes

- Practical knowledge of loading, inspecting and cleaning data with Pandas.
- Ability to calculate business metrics such as revenue and average order value.
- Experience in grouping data and comparing products, regions and months.
- Skill in creating bar charts and line charts with Matplotlib.
- Understanding of why validation and clear documentation are needed before conclusions are drawn.

12. FUTURE WORK

The present work can be extended in the following ways. Table 12.1 lists each enhancement together with the benefit it would bring.

Enhancement	Benefit
Loading sales data directly from a MySQL database instead of CSV files	Removes the need to export files and keeps the analysis connected to live records
Processing multiple CSV and Excel files and scheduling the analysis to run automatically	Saves manual effort when new files arrive regularly
Creating interactive dashboards	Allows easier exploration of product, region and monthly sales
Adding sales forecasting using time series and machine learning techniques	Helps in planning stock and promotions in advance
Extending the analysis to customer-wise and category-wise sales and to profit analysis	Gives a fuller picture of business performance
Developing a web application where users can upload a sales file and view the results online	Makes the analysis available to users without programming knowledge
Generating the final report automatically as PDF or Excel files	Reduces the time spent on preparing reports
Adding detection of unusual values and alerts for invalid records	Improves data quality by flagging problems early

Table 12.1: Possible future enhancements

APPENDIX A: DATA QUALITY AND VALIDATION

Data quality is a major factor in the reliability of a sales analysis report. Before calculating revenue, the analyst should check whether the records are complete, consistent, valid and suitable for the intended purpose. A dataset may contain missing values, duplicate orders, spelling variations, invalid dates or numbers stored as text. If these problems are ignored, the summary tables and charts may give a misleading picture of business performance.

The first check is completeness. The analyst reviews the number of empty cells in each column and identifies whether the missing value affects the analysis. For example, a missing quantity prevents a direct revenue calculation, while a missing region prevents the order from being included in a regional comparison. The handling method should be recorded so that the decision can be reviewed later.

The second check is consistency. Product and region names should use a common format. Extra spaces and inconsistent letter cases can cause the same category to appear as multiple categories in a grouped summary. Date values should also follow a consistent format before monthly grouping is performed.

The third check is validity. Quantity and unit price should be numeric, and unexpected negative or zero values should be investigated according to the business rules. Duplicate checks should use appropriate identifying fields and not remove valid repeat purchases simply because two rows look similar.

Finally, the analyst should validate totals after cleaning. The sum of product revenue, region revenue and monthly revenue should agree with total revenue when the groupings cover the same valid records. These checks increase confidence in the report and make the analysis easier to reproduce.

APPENDIX B: EXPLANATION OF THE PYTHON LIBRARIES

Python is used as the main programming language because its syntax is readable and it provides a wide range of libraries for data analysis. The project uses Pandas, NumPy, Matplotlib and Seaborn to perform different parts of the workflow. Each library has a specific purpose, and together they reduce the amount of repetitive code required for analysis.

Pandas is used to read the CSV file and store the records in a DataFrame. A DataFrame arranges data in rows and columns, similar to a spreadsheet. Pandas provides methods to inspect the dataset, identify missing values, remove duplicate rows, convert dates, group records and calculate totals. For this project, it is the main library for data preparation and aggregation.

NumPy supports numerical operations and works closely with Pandas. It provides functions for numerical calculations and array-based operations. Statistical values such as medians can be used during data preparation, while numerical checks help the analyst review the results.

Matplotlib is used to create and save charts. A bar chart is appropriate for comparing product or regional revenue because the height of each bar represents the value for a category. A line chart is useful for monthly analysis because the connected points make changes over time easier to see. Clear titles, axis labels and suitable scales improve readability.

Seaborn provides a higher-level interface for statistical visualizations and can help produce consistent chart styles. Even when Matplotlib is used directly, Seaborn remains a useful option for future improvements. The selected tools are widely used in introductory data analysis and allow the workflow to be extended as project requirements grow.

APPENDIX C: BUSINESS INTERPRETATION OF SALES RESULTS

The purpose of sales analysis is not only to calculate totals but also to interpret what the numbers may mean for a business. In the sample results, Smartphone accounts for the largest share of revenue. This indicates that smartphones are an important contributor within the sample dataset. A business could use this observation as a reason to review stock availability, product demand and supplier planning. However, revenue alone does not establish profitability because purchase costs and operating expenses are not included.

The North region has the largest regional contribution in the sample. Regional differences may be related to customer demand, distribution, local competition, marketing activity or availability. The dataset does not contain enough information to identify the cause. The appropriate next step would be to collect additional data, such as the number of customers, number of orders, promotional activity and stock availability for each region.

Monthly sales are lowest in April and highest in June in the sample period. This change may help a business decide when to review inventory and promotional schedules. However, only six months of data are available, so it would be premature to describe the pattern as a confirmed annual seasonal trend. More months and multiple years of observations would provide a stronger basis for such a conclusion.

A responsible interpretation separates facts from hypotheses. A fact is directly supported by the calculation, such as the month with the highest revenue. A hypothesis is a possible explanation that needs additional evidence. This distinction prevents the report from making claims that are stronger than the data supports. Business recommendations should therefore be treated as suggestions for further investigation rather than automatic decisions.

APPENDIX D: TESTING AND ERROR HANDLING

Testing checks whether the program behaves as intended and whether its outputs are internally consistent. The Sales Data Analysis workflow can be tested by checking each module separately and then checking the complete script. Module-level testing makes it easier to locate errors because each stage has a defined input and output.

For data loading, the program should confirm that the expected columns are present. If the file path is incorrect or a required column is missing, the program should display a clear error rather than continue with incomplete information. Data type checks should confirm that dates can be converted and that quantity and unit price can be used in arithmetic operations.

For cleaning, test cases should include a duplicate row, a missing quantity, a missing region, and category names containing extra spaces or different letter cases. The expected result should be decided before running the test. For example, a duplicate record should be handled according to the defined duplicate rule, and a missing region should not silently appear as a valid regional category.

For revenue calculation, a small hand-checked example is useful. If an order contains a quantity of two and a unit price of INR 100, the calculated revenue should be INR 200. This simple test checks whether the multiplication and field references are correct.

For aggregation, compare the product-wise, region-wise and month-wise totals with the overall revenue. If any total differs, inspect missing categories, invalid dates and records excluded during grouping. Testing should also include a review of generated chart files to confirm that they are saved and have meaningful titles and labels. Recording test cases and outcomes makes the program easier to maintain.

APPENDIX E: DATA PRIVACY AND RESPONSIBLE REPORTING

Sales records may contain commercially sensitive information. Even when a dataset does not include customer names, transaction patterns, product pricing and regional performance may still be confidential. Analysts should use only data they are authorised to access and should follow the organisation's rules for storing, sharing and retaining files.

A report intended for academic submission should clearly identify whether its figures come from verified company records or illustrative sample data. Sample values must not be presented as actual company results. This report uses illustrative sample figures to explain the analysis workflow; they should be replaced with approved company data only when such data is available and permission has been granted.

Files should be stored in appropriate folders, and unnecessary copies should be avoided. If the dataset contains personal or sensitive fields, those fields should not be included in charts or tables unless they are required and authorised. When sharing results, aggregated figures are often sufficient for explaining trends without exposing individual transactions.

Responsible reporting also requires accurate wording. The analyst should describe the period covered, explain important data-cleaning decisions and mention limitations that may affect interpretation. A visualisation should not be designed to exaggerate small differences or hide relevant context. Axis labels, units and chart titles should be clear.

These practices help preserve trust in the analysis. Data quality, privacy and transparent reporting are not separate from technical work; they are important parts of producing useful and credible analytical results.

APPENDIX F: PROJECT CHALLENGES AND LEARNING

One challenge in a sales analysis project is that raw data is rarely perfectly organised. Values may be missing, categories may be written in different formats, and duplicate records may affect totals. Learning to inspect the data before performing calculations is therefore an important part of the workflow. The project demonstrates that data preparation is not simply a preliminary task; it directly affects the quality of the final results.

Another challenge is choosing an appropriate method to summarise the information. A total revenue figure provides an overall view, but it does not explain which products or regions contribute most. Grouping the data by product, region and month provides different perspectives. Visualisation complements the tables by allowing patterns and differences to be recognised quickly.

The project also highlights the importance of checking results. A script can run without producing a programming error and still contain a logical mistake. Comparing grouped totals with overall revenue is one example of a validation check. Reviewing chart labels and the date range provides additional assurance that the results have been presented correctly.

Through this work, practical skills are developed in reading files, working with tabular data, handling missing values, calculating derived fields, grouping records and creating charts. The project also strengthens problem-solving skills because each stage depends on the quality of the previous stage.

The experience can support further learning in SQL, dashboard development, database connectivity and forecasting. These topics build on the same foundation: understanding the data, preparing it carefully, analysing it with suitable methods and communicating the results clearly. The project provides a starting point for more advanced data analytics work.

APPENDIX G: RECOMMENDED WORKFLOW FOR FUTURE DATA UPDATES

When a new sales file becomes available, the same workflow can be reused with a few checks. First, keep an unchanged copy of the original file so that the source data can be reviewed if an unexpected result appears. Confirm the file name, file location and column names before running the script. If the structure has changed, update the loading or preparation logic deliberately rather than silently ignoring the difference.

Next, inspect the new file. Review the number of rows, date range, missing values, duplicate records and numeric fields. Compare these checks with the previous run to identify unusual changes. For example, a sudden increase in missing unit prices may indicate a source-system problem rather than a genuine business pattern.

After cleaning, calculate revenue and produce the product, region and monthly summaries. Validate that the grouped totals reconcile with total revenue. Save the charts and summary tables with filenames that clearly identify the analysis period. If the report is generated repeatedly, include the run date and the period covered in the report metadata.

Before sharing the output, review the findings and confirm that every statement is supported by the tables or charts. Avoid comparing periods of different lengths without explanation. If the source data has changed, document the change and explain how it may affect the comparison.

A future version of the project could automate these checks, create a dashboard and export the results to PDF or Excel. The basic principles would remain the same: preserve the source, validate the data, apply consistent calculations, check the results and communicate the limitations clearly.

13. REFERENCES

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